1. Project Purpose

This project is a role-based airline management demo application built with **Python**, **SQLite**, and **Tkinter**.
Its main goal is to automatically initialize the database from `setup.sql` on first run, load seed data, and start a desktop interface with two separate role flows.

The system demonstrates realistic role separation:

- **Passenger**: search flights and create reservation requests
- **Admin**: manage flights and approve/reject requests

2. Technologies Used

- **Python 3** – core application logic
- **SQLite** – lightweight file-based database
- **Tkinter** – desktop GUI

SQLite was chosen for simple setup (no server needed), and Tkinter for fast UI development without extra dependencies.

3. Database Initialization

When the app starts, it checks for `airline.db`.
If the database or required tables are missing, `setup.sql` runs automatically and creates schema + seed data.

Main structures:

- `users`
- `flights`
- `reservation_requests`
- `v_flight_summary` (view)
- `v_request_overview` (view)

Business rules are enforced with triggers (e.g., no duplicate seat on the same flight, no request when capacity is full).

4. Roles and Workflows

Passenger

Passengers can:

- filter and list flights

- create reservation requests with seat number and note
- track their request statuses
- cancel pending requests

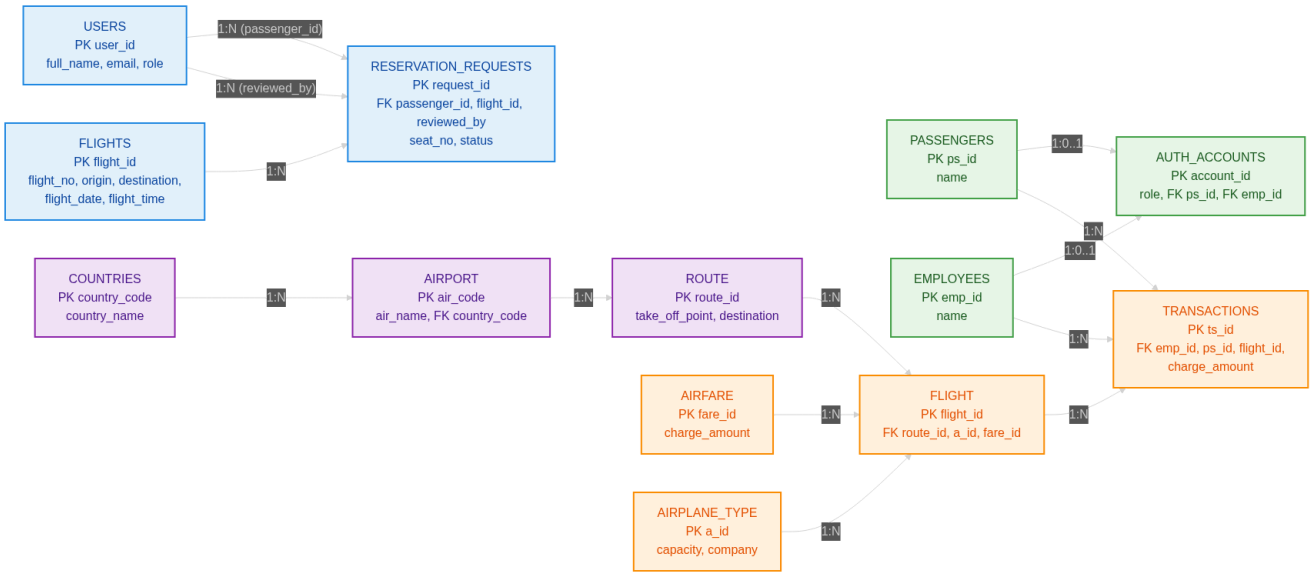
Admin

Admins can:

- add, update, delete flights
- filter and review reservation requests
- approve, reject, or cancel requests
- create demo users

This separation is the core requirement of the project.

5. ER Model Summary


USERS
PK user_id
full_name, email, role
1:N (passenger_id)
1:N (reviewed_by)
RESERVATION_REQUESTS
PK request_id
FK passenger_id, flight_id, reviewed_by
seat_no, status
FLIGHTS
PK flight_id
flight_no, origin, destination, flight_date, flight_time
1:N
PASSENGERS
PK ps_id
name
1:0..1
AUTH_ACCOUNTS
PK account_id
role, FK ps_id, FK emp_id
1:N
1:0..1
COUNTRIES
PK country_code
country_name
1:N
AIRPORT
PK air_code
air_name, FK country_code
1:N
ROUTE
PK route_id
take_off_point, destination
1:N
EMPLOYEES
PK emp_id
name
1:N
TRANSACTIONS
PK ts_id
FK emp_id, ps_id, flight_id, charge_amount
AIRFARE
PK fare_id
charge_amount
1:N
FLIGHT
PK flight_id
FK route_id, a_id, fare_id
1:N
AIRPLANE_TYPE
PK a_id
capacity, company
1:N

flights
INTEGER flight_id PK
TEXT flight_no UK
TEXT origin
TEXT destination
TEXT flight_date
TEXT flight_time
INTEGER capacity
users
INTEGER user_id PK
TEXT full_name
TEXT email UK
TEXT role
TEXT created_at
Countries
TEXT Country_code PK
TEXT Country_Name UK
reservation_requests
INTEGER request_id PK
INTEGER passenger_id FK
INTEGER flight_id FK
TEXT seat_no
TEXT status
TEXT note
INTEGER reviewed_by FK
TEXT created_at
TEXT decided_at
Airport
TEXT Air_Code PK
TEXT Air_Name
TEXT City
TEXT State
TEXT Country_code FK
AirFare
INTEGER Fare_ID PK
REAL Charge_Amount UK
TEXT Description
Route
INTEGER Route_ID PK
TEXT Destination FK
TEXT Take_Off_point FK
TEXT R_type
Airplane_type
INTEGER A_ID PK
INTEGER Capacity
REAL A_weight
TEXT Company
Flight
INTEGER Flight_ID PK
TEXT Flight_No UK
TEXT Departure
TEXT Arrival
TEXT Flight_date
INTEGER Route_ID FK
INTEGER A_ID FK
INTEGER Fare_ID FK
Employees
INTEGER Emp_ID PK
TEXT Name
TEXT Address
INTEGER Age
TEXT Email_ID UK
TEXT Contacts
TEXT Air_Code FK
Passengers
INTEGER Ps_ID PK
TEXT Name
TEXT Address
INTEGER Age
TEXT Sex
TEXT Contacts
Travels_on
INTEGER Route_ID PK,FK
INTEGER Flight_ID PK,FK
Transactions
INTEGER TS_ID PK
TEXT Booking_Date
TEXT Departure_Date
TEXT Type
INTEGER Emp_ID FK
INTEGER Ps_ID FK
INTEGER Flight_ID FK
REAL Charge_Amount FK
Can_Land
TEXT Air_Code PK,FK
INTEGER Flight_ID PK,FK
Auth_Accounts
INTEGER Account_ID PK
TEXT Role
INTEGER Ps_ID FK
INTEGER Emp_ID FK
TEXT Password_Hash
TEXT Created_At